

Thông tin sinh viên

* Indicates required question

1. Email *

2. 1. Mã sinh viên (In hoa, ví dụ: N23DCAT015). *

Chú ý ghi đúng mã sinh viên để tính điểm, ghi sai sẽ không có điểm

3. 2. Họ và tên (Ví dụ: Nguyễn Văn A) *

4. 3. Lớp (Ví dụ: D23DCAT01-N) *

QUIZ

5. What information does a Stateless Firewall rely on to decide whether to block or allow a packet? *

Mark only one oval.

- ☐ The content of executable HTML code and database return messages inside the Payload.
- ☐ The source/destination IP address and transport protocol port (Layer 3 & 4) recorded on the Header.
- ☐ Changes in the Entropy level of signal streams according to the intrusion detection model.
- ☐ The validity of the SSL certificate and public key encryption algorithm on the web.

6. How does a Stateful Firewall overcome the weakness of a stateless firewall? *

Mark only one oval.

- ☐ By automatically encrypting all internal access streams to protect file safety.
- ☐ By tracking the TCP handshake process to dynamically open ports for valid response packets.
- ☐ By scanning and killing viruses directly within text attachments in email services.
- ☐ By preventing employees from copying company source code to portable hard drives.

7. Which characteristic allows an Application-Level Gateway to block SQL Injection attacks? *

Mark only one oval.

- ☐ Monitoring source IPs belonging to blacklists of international cybercrime organizations.
- ☐ Operating at OSI Layer 7, capable of reading and understanding SQL commands deep inside the packet.
- ☐ Integrating broadband routing hardware to neutralize denial of service attacks.
- ☐ Automatically changing DNS ports to mislead vulnerability scanning applications.

8. How does a Circuit-Level Gateway primarily determine a valid connection session? *

Mark only one oval.

- ☐ By inspecting the connection establishment handshake process (e.g., TCP Three-way handshake).
- ☐ By filtering unusual keywords appearing in HTTP headers or webpage content.
- ☐ By requiring administrators to use biometric fingerprints to authorize each data packet.
- ☐ By analyzing WiFi signal strength to exclude unauthorized remote access devices.

9. Why do Internet Service Providers (ISPs) prefer using Stateless Firewalls in the core zone instead of more secure Application Firewalls? *

Mark only one oval.

- ☐ Because packet filtering firewalls have complex algorithms that completely block phishing email campaigns.
- ☐ Because processing the application layer consumes too many resources, causing severe network latency for massive traffic.
- ☐ Because international organizations do not sell Application Firewall licenses for telecom systems.
- ☐ Because packet filtering firewalls have interfaces allowing customers to change transmission parameters unassisted.

10. If a rule on a Stateless Firewall is set to allow LAN computers to access the Internet via port 443, what else must an engineer do to enable web browsing? *

Mark only one oval.

- ☐ Enable advanced anti-malware features to scan images downloaded from the internal network.
- ☐ Write an Outbound rule forcing the web server to encrypt documents with public digital certificates.
- ☐ Open an additional Inbound rule allowing response data streams from the Internet to enter the LAN's dynamic ports.
- ☐ Lock DNS routing streams to restrict spyware from tracking web browsing history.

11. The UDP protocol has no connection handshake process. How does a Stateful Firewall manage UDP streams from inside to outside? *

Mark only one oval.

- ☐ Permanently blocks all UDP requests, evaluating it as an insecure protocol standard.
- ☐ Saves the source/destination IP and Port in a temporary memory; if a returning UDP packet matches the parameters, the firewall lets it pass.
- ☐ Requires the destination server to send an RSA encryption certificate to prove the packet's origin before passing.
- ☐ Forces the application to switch from UDP to TCP before being granted Internet access.

12. A Web Application Firewall (WAF) is an advanced version of which type of firewall? *

Mark only one oval.

- ☐ Circuit-Level Gateway.
- ☐ Packet-Filtering Router.
- ☐ Application-Level Gateway analyzing detailed Payload data.
- ☐ Host-based IDS platform collecting server login logs.

13. An accountant accidentally clicks a link downloading malware via HTTPS. Even though the Stateful Firewall is configured * to block all Inbound traffic, the malware still gets through. Why?

Mark only one oval.

- ☐ The malware uses MAC address spoofing techniques causing the firewall to misidentify internal network devices.
- ☐ This file download process is the response data stream of an HTTPS connection proactively initiated by the internal computer.
- ☐ A stateful firewall defaults to ignoring and not inspecting any packets running on the UDP protocol.
- ☐ The administrator forgot to update the intrusion detection signature into the central router's memory.

14. The Slowloris attack trickles HTTP Headers to hang Web servers. Why is a Stateful Firewall ineffective against this attack * type?

Mark only one oval.

- ☐ The firewall defaults to blocking all connections with data sending speeds lower than hardware specifications.
- ☐ Because the TCP handshake is entirely valid, the firewall considers this a normal connection and keeps the port open.
- ☐ HTTP is a highly secure application so the firewall is not permitted to inspect packet parameters.
- ☐ The attacker hijacked DNS and modified the routing table of the internal incident monitoring system.

15. A hacker fragments an exploit code into ultra-small IP Fragments and sends them out of order. What is the weakness of a Stateless Firewall against this attack method? *

Mark only one oval.

- ☐ It immediately suspends all Internet-originating connections upon detecting non-standard fragmentation.
- ☐ A static firewall evaluates each packet individually, lacking a buffer to reassemble the fragments and review the entire content.
- ☐ A static firewall automatically converts fragments into images to conduct biometric analysis for integrity.
- ☐ It requires employees to enter a confirmation code before authorizing the redistribution of data fragments to the server.

16. A hacker uses Nmap for port scanning and receives a "Filtered" result instead of "Open" or "Closed". How did the stateful firewall process this probe packet? *

Mark only one oval.

- ☐ It returned malware directly targeting the scanning computer to destroy its database.
- ☐ It directly dropped the probe packet silently, without returning an ICMP error response, to hide server information.
- ☐ It let the packet pass but redirected the traffic to a wide-area intrusion log analysis center.
- ☐ It replaced TCP with UDP to neutralize Nmap's bandwidth probing algorithm due to misconfiguration.

17. A firewall has a rule: "Block all .exe files". An attacker renames the malware file to `trojan.jpg`. Why does an Application-Level Gateway still detect and block it? *

Mark only one oval.

- ☐ It uses file surface analysis technology cross-referenced with network user surveillance camera profiles.
- ☐ It looks beyond the fake extension (.jpg) and analyzes deep structures (Magic bytes) to identify the true format.
- ☐ It establishes a direct query line with security agencies to determine the IP distributing the image file.
- ☐ It features a memory of users who sent fake files in the past and automatically blocks based on prediction.

18. A firewall only allows HTTPS (port 443) and forbids FTP (port 21). What control technique is this? *

Mark only one oval.

- ☐ Direction Control technique.
- ☐ Service Control technique.
- ☐ User Control technique.
- ☐ Behavior Control technique.

19. A firewall blocks Internet connections directed into the internal network, but allows all computers in the internal network to connect out to the Internet. Which technique is being used? *

Mark only one oval.

- ☐ Service control by closing malicious web applications.
- ☐ Behavior control by analyzing network programming languages.
- ☐ User control via facial recognition systems.
- ☐ Direction control by managing the connection stream's initiation direction.

20. The company network is congested due to employees downloading Torrent files. To solve this using the "Behavior Control" technique on the Firewall, what should the administrator do? *

Mark only one oval.

- ☐ Block all outgoing transmission connections from the enterprise network during business hours.
- ☐ Use P2P Torrent protocol signature recognition algorithms to block or throttle this bandwidth range.
- ☐ Force employees to scan fingerprints whenever they want to download any file from the network.
- ☐ Close all Web server ports (80, 443) to prevent users from browsing the web to download files.

21. Requiring remote employees to log into a VPN with an account and OTP code before accessing data is an example of which firewall technique? *

Mark only one oval.

- ☐ Content filtering technique for email attachments.
- ☐ Transmission speed limiting technique for social media platforms.
- ☐ User Control technique to grant resource access rights.
- ☐ Technique redirecting suspicious traffic into Honeypots.

22. To defend against IP Spoofing where an Internet hacker fakes a source IP from the LAN range (e.g., 192.168.1.x) to bypass the firewall, what rule must be set? *

Mark only one oval.

- ☐ Allow all packets with source IP 192.168.1.x originating from the Internet port to enter the network.
- ☐ Immediately deny any data packet entering the firewall from the Internet port but carrying a LAN source IP address.
- ☐ Install a smartcard authentication system for all packets carrying a LAN IP accessed from the outside.
- ☐ Forward web application ports to the DMZ so the database server automatically identifies intruders.

23. Firewall Rules: Rule 1: Allow TCP LAN to Any:443; Rule 2: Deny Any to Any. How will a UDP packet sent from the internal network to an international DNS server over port 53 be processed? *

Mark only one oval.

- ☐ Permitted because the DNS protocol has firewall bypass capabilities.
- ☐ Blocked by Rule 2 because it does not meet Rule 1's TCP and port 443 conditions.
- ☐ Permitted because Rule 1 implicitly allows all outbound connections from the LAN.
- ☐ Suspended pending administrator verification for malicious content.

24. If setting up a firewall to control access by Domain Name instead of IP address, what risk is the system prone to? *

Mark only one oval.

- ☐ Firewall overload due to storing high-quality domain name images in temporary memory.
- ☐ Hackers launching a DNS cache poisoning attack, tricking the firewall into allowing access to malicious servers.
- ☐ DAC authorization automatically changing top-secret data labels upon receiving a domain request.
- ☐ Continuous decryption of the SSL protocol causing financial data leaks externally that IDS cannot control.

25. What is the risk of setting the firewall rule "Allow all Inbound and Outbound connections for the CEO's computer IP address"? *

Mark only one oval.

- ☐ If the CEO's computer gets infected with malware, the malware will exploit this privilege to easily spread directly into the core network zone.
- ☐ Firewall performance drops sharply due to using all CPU power encrypting packets sent from the server to this computer.
- ☐ Setting priority for one computer makes the system unable to create any other access control lists.
- ☐ The CEO's computer gets disconnected constantly when the firewall downloads malware signature updates from the Internet.

26. Among the following security risks, which behavior falls outside the prevention capabilities of a Perimeter Firewall? *

Mark only one oval.

- ☐ Hackers sending millions of packets to overload the domain name resolution port.
- ☐ An internal employee using a USB to copy the entire database and take it home.
- ☐ An attacker scanning open and responding web service ports.
- ☐ Requests to update software downloaded from Internet app stores.

27. Why does a standard packet-filtering firewall often fail to detect malware hidden inside HTTPS or SSL/TLS connections? *

Mark only one oval.

- ☐ The firewall lacks a GUI displaying warnings to SOC administrators.
- ☐ The packet-filtering firewall cannot decrypt the content, hence cannot read the actual Payload data.
- ☐ The SSL certificate provider server blocks the routing access process of the packet inspection device.
- ☐ The firewall defaults to allowing secure protocol ports open without controlling connection directions.

28. How does Network Segmentation overcome the weakness of a "single perimeter firewall" architecture? *

Mark only one oval.

- ☐ Groups network devices into a single inspection loop for easier internal malware tracking.
- ☐ Divides the large LAN into multiple isolated zones with individual internal firewalls, limiting the lateral spread path of malware.
- ☐ Encrypts all server data into static files, eliminating decompression capabilities via firewall devices.
- ☐ Replaces user management with biometrics integrated with OTP to reduce routing buffer loads.

29. The "SSL Inspection" application decrypts, virus-scans, and re-encrypts HTTPS packets right at the firewall. What is the biggest challenge when deploying this technology? *

Mark only one oval.

- ☐ Requires replacing the entire old network cable system with the newest satellite transmission lines.
- ☐ Alters user hard drive formats, causing data loss when the OS automatically updates.
- ☐ Consumes massive amounts of the firewall's CPU resources and raises legal risks regarding privacy violations.
- ☐ False digital certificate warnings will jam the system, rejecting all company internal app accesses.

30. When moving infrastructure to the virtualized cloud, servers communicate internally on an "East-West" axis. Why can't a perimeter firewall protect this stream? *

Mark only one oval.

- ☐ East-West traffic flows directly between internal virtual machines, not passing through the firewall placed at the network perimeter.
- ☐ Perimeter firewalls are specifically designed to block mobile digital certificates, not protect cloud VM transmissions.
- ☐ Cloud routing automatically rejects packets lacking physical MAC address verification from workstation processors.
- ☐ Virtualized space uses biometric verification, making physical firewalls incompatible with the sensors.

31. In Zero Trust architecture, how does the "Verify every request" principle change the access control process compared to traditional firewalls? *

Mark only one oval.

- ☐ Identity verification and authorization are performed per application session, no longer defaulting to trusting devices inside the internal LAN.
- ☐ NIDS systems switch to network anomaly analysis to retrieve cloud server signature samples across regions.
- ☐ Firewalls must block international IPs and disable all admin rights because devices are app-locked.
- ☐ All multimedia file sharing actions must be pre-approved by administrators to eliminate malware in images.

32. Hackers exploit the DNS service port (UDP 53) legitimately opened on the firewall to secretly transmit command and control payloads. What reason makes the defense system completely bypassed? *

Mark only one oval.

- ☐ The firewall does not block port 53; meanwhile, the MAC model cannot prevent external routing streams from hijacking internal servers.
- ☐ The firewall auto-redirects port 53 connections to web 443; the DAC model accidentally sends fake digital certificates to admin emails.
- ☐ The stateful firewall drops the connection tracking table, cutting off IPS's information source for analyzing bandwidth violation logs.
- ☐ The firewall perfectly encrypts exploit packets, helping them slip past biometric scanners and alter cloud storage hashes.

33. What is the most optimal solution to overcome the perimeter firewall's weakness of "not detecting Advanced Persistent Threats (APT)"? *

Mark only one oval.

- ☐ Force employees to change passwords weekly and use USB-blocking software against data copying.
- ☐ Upgrade to the WPA3 wireless standard for the entire branch office network.
- ☐ Use a Security Information and Event Management (SIEM) system combined with IDS to analyze and link prolonged minor behaviors into an attack chain.
- ☐ Build a web application firewall (WAF) filter to eliminate computer malware infections via fake images.

34. A company uses WAF and HIDS to protect a web application server. A hacker detects the SSH admin port (Port 22) open because the admin forgot to block it on the perimeter firewall. What is the consequence of this error? *

Mark only one oval.

- ☐ WAF only protects web applications (HTTP/HTTPS); leaving SSH (Layer 4) open helps hackers bypass the WAF barrier and access the OS.
- ☐ The server's digital signature has been altered by hackers, making the workstation config connection port invisible to scanners.
- ☐ The perimeter firewall automatically disconnects traffic when public key encryption conflicts occur, exposing accountant account data.
- ☐ The HIDS platform on the server is not granted permission to scan for storage structure changes related to virtual telecommunications software.

35. What is the primary functional difference making an Intrusion Prevention System (IPS) distinct from an Intrusion Detection System (IDS)? *

Mark only one oval.

- ☐ The ability to save detailed computer logs to serve in-depth security investigation processes.
- ☐ The ability to analyze and automatically take immediate action to block malicious packets in real time.
- ☐ The ability to passively monitor multidimensional data streams between core and office networks using mirror ports.
- ☐ The ability to use massive signature databases to scan for ransomware formats attached to cloud emails.

36. How do IDS sensors typically collect out-of-band traffic data? *

Mark only one oval.

- ☐ By directly splicing into the main fiber optic backbone located between the router and the DMZ firewall.
- ☐ By installing tracking apps on every workstation and phone to analyze app integrity.
- ☐ By configuring Port Mirroring/SPAN features on Switches to send a copy of network traffic to the IDS.
- ☐ By setting the sensor as the sole routing point to verify the validity of all incoming and outgoing Internet connections.

37. Deploying an IPS to block data streams directly (Inline mode) creates what major system risk? *

Mark only one oval.

- ☐ IPS completely loses the ability to detect sophisticated malware due to hardware-limited routing capacity.
- ☐ If the IPS is overloaded or its hardware hangs, it becomes a Single Point of Failure (SPOF) paralyzing all network connectivity.
- ☐ Internal databases are accidentally pushed out to the Internet zone due to secure comms data stream branching configuration.
- ☐ The firewall encryption inspection feature gets manually turned off to avoid rule conflicts between admin systems and workstations.

38. A company configures IPS with excessively high sensitivity. Consequently, the IPS automatically severs thousands of valid customer transaction sessions. What is this scenario called? *

Mark only one oval.

- ☐ Buffer Overflow attack causing slow server responses and misdirected queries.
- ☐ False Positive causing mistakenly blocked legitimate traffic, disrupting business processes.
- ☐ False Negative allowing targeted attacks to slip into the core server zone.
- ☐ Routing software bandwidth overload causing the firewall to shuffle permission tables into censorship lists.

39. For a telecom core network with thousands of Gigabits/second traffic requiring low latency, why use passive IDS instead of inline IPS? *

Mark only one oval.

- ☐ Passive IDS only processes on data copies, causing no bandwidth hindrance or network drop risks to the original transaction stream.
- ☐ IDS devices automatically identify and immediately apply OS patches via distributed secure protocol channels.
- ☐ IDS algorithms possess the power to stop direct denial of service attacks against web server clusters without consuming CPU.
- ☐ IDS proactively fully decrypts all data streams through the standard digital certificate system encryption structure.

40. When an IDS sensor detects a packet containing a critical vulnerability exploit signature, what does it automatically do? *

Mark only one oval.

- ☐ Sends a recovery encryption file to the compromised victim computer for urgent patching.
- ☐ Logs the violating packet into security logs and immediately alerts the central SOC administrator.
- ☐ Cuts power breakers supplying perimeter routers connected directly to the spoofed address.
- ☐ Purges the malicious code stream from the cache of the app server OS currently suffering the attack.

41. An administrator switches an analysis device from auto-blocking packets to a mode that only identifies and issues on-screen alerts to avoid internal workflow disruption. This action is equivalent to what? *

Mark only one oval.

- ☐ Downgrading from a secure Virtual Local Area Network to a low-level fragmented routing model.
- ☐ Downgrading from an active Web Application Firewall back to a static packet filter.
- ☐ Downgrading from an inline anti-malware system to a DMZ honeypot decoy system.
- ☐ Downgrading from an Intrusion Prevention System (IPS) back to a passive Intrusion Detection System (IDS).

42. What is an analytical platform installed directly onto a server's operating system to monitor file changes and system login logs called? *

Mark only one oval.

- ☐ Stateless packet filtering firewall system operating based on network port connection routing.
- ☐ Network Intrusion Detection System (NIDS) analyzing electronic info streams on fiber optics.
- ☐ Integrated security stream distribution system preventing background connections at internal network addresses.
- ☐ Host-based Intrusion Detection System (HIDS) protecting the operational structure of a specific device.

43. NIDS encounters difficulties ("blind spots") in an enterprise Switch environment if administrators do not implement which configuration? *

Mark only one oval.

- ☐ Updating the SSL digital certificate encryption license from a multinational web-protection CA.
- ☐ Setting up Port Mirroring/SPAN so NIDS receives traffic copies between workstations plugged into the same Switch.
- ☐ Turning off the Windows Defender firewall feature on workstations so NIDS can scan for OS behavior anomalies.
- ☐ Changing the operating system's open-source programming language to a licensed Windows server.

44. A hacker uses a USB to insert malware directly infecting the accounting server (bypassing the LAN). The perimeter firewall and NIDS are both neutralized. Which tool detects this incident? *

Mark only one oval.

- ☐ Host-based Intrusion Detection System (HIDS) monitoring abnormal changes in the file system on the machine.
- ☐ Distributed Intrusion Prevention System automatically patching routing database software.
- ☐ Circuit-level gateway established on the accounting server's virtual port.
- ☐ Stateful firewall system capable of analyzing cache signatures of HTTPS protocol data streams.

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